

## IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

### PROJECT 35

# Enable Session Manager Logging to S3

AWS Systems Manager | Session Manager | Amazon S3 | IAM | EC2

|              |                  |
|--------------|------------------|
| AUTHOR       | Michael Salazar  |
| ENVIRONMENT  | Personal AWS Lab |
| VERSION      | v1.0.0           |
| RELEASE DATE | July 23, 2026    |
| STATUS       | Complete         |

## Executive Summary

Logged into **TestAccount1** with full administrator access, created a new **SLAW** CloudFormation stack from the provided template URL, enabled **Session Manager S3 logging**, selected the **session-manager-logs** S3 bucket, and updated the **SSMClient** IAM role with a customer inline policy named **SSMLogs**. The inline policy allowed the managed instance to write Session Manager activity logs to the selected S3 bucket. After connecting to the instance, running commands, exiting the session, and confirming the downloaded log file, the lab was cleaned up by emptying the log bucket, removing the inline policy, and deleting the CloudFormation stack.

### SECURITY OUTCOME

This project adds command accountability to private EC2 access. Session Manager already avoids SSH exposure; S3 logging preserves a reviewable record of commands typed during the session.

## Business Problem

Previous projects demonstrated private EC2 access with Session Manager, VPC endpoints, Run As, and CLI-based access. That access path still needs audit evidence for incident response and accountability. This lab enables Session Manager activity logging to S3 so interactive commands can be reviewed after the session ends.

## Objectives

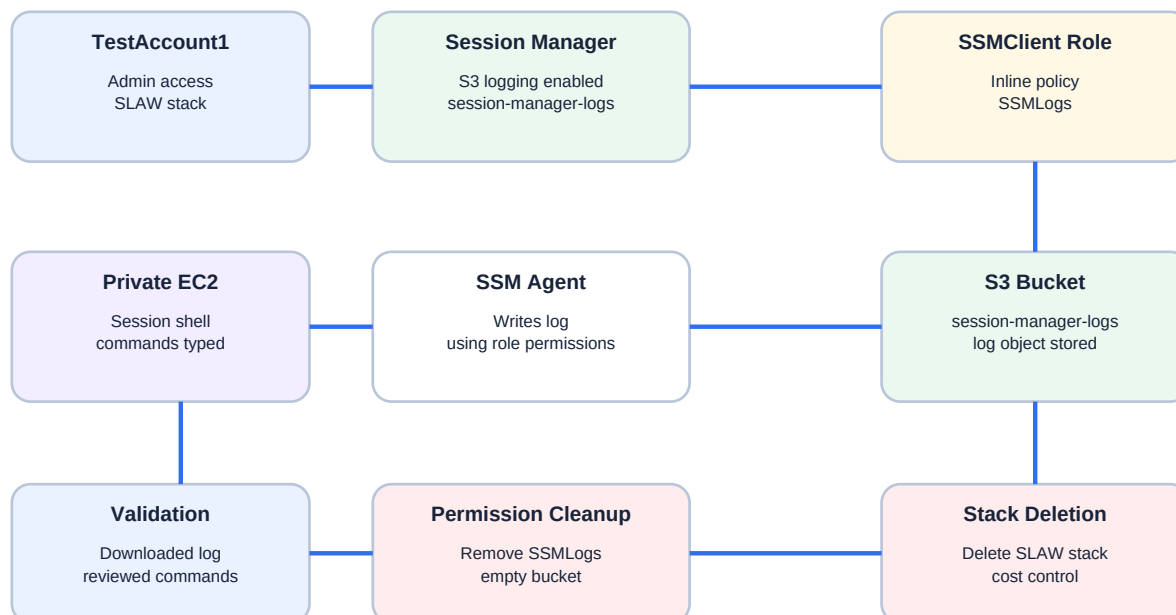
- Sign into TestAccount1 with AdministratorAccess.
- Create the SLAW CloudFormation stack from the provided lab template URL.
- Enable Session Manager S3 logging.
- Select the session-manager-logs S3 bucket as the log destination.
- Open IAM Roles and update the SSMClient role.
- Create a customer inline policy named SSMLogs.
- Allow s3:PutObject to the selected bucket path and s3:GetEncryptionConfiguration for encryption validation.
- Connect to the EC2 instance through Session Manager, run sample commands, and exit.
- Confirm the S3 log object was created and download the log for review.
- Empty the logging bucket, remove the inline policy, and delete the stack for cleanup.

## Environment

| Cloud Provider   | Amazon Web Services (AWS)                                                               |
|------------------|-----------------------------------------------------------------------------------------|
| Primary Services | AWS Systems Manager Session Manager, Amazon S3, AWS IAM, Amazon EC2, AWS CloudFormation |
| Account Context  | TestAccount1 / AdministratorAccess                                                      |

| Stack Name                 | SLAW                                                         |
|----------------------------|--------------------------------------------------------------|
| Template Source            | Provided CloudFormation template URL from the lab            |
| Session Log Destination    | S3 bucket beginning with session-manager-logs                |
| Session Manager Preference | S3 logging enabled                                           |
| IAM Role Updated           | SSMClient                                                    |
| Inline Policy              | SSMLogs                                                      |
| Policy Actions             | s3:PutObject and s3:GetEncryptionConfiguration               |
| Validation                 | Downloaded Session Manager log captured command activity     |
| Cleanup                    | Emptied log bucket, removed inline policy, and deleted stack |

## Architecture



Session Manager preferences set the log destination; the EC2 role lets the SSM Agent write the S3 object.

Figure 1 - Session Manager S3 logging workflow.

## Implementation Summary

1. Signed into TestAccount1 using full administrator access.
2. Created the SLAW CloudFormation stack using the provided lab template URL.

3. Opened Systems Manager and navigated to Session Manager Preferences.
4. Enabled S3 logging and selected the session-manager-logs bucket.
5. Opened IAM and navigated to Roles.
6. Opened the SSMClient role used by the managed EC2 instance.
7. Created a customer inline policy named SSMLogs.
8. Configured the inline policy to allow s3:PutObject to the session-manager-logs bucket path and s3:GetEncryptionConfiguration.
9. Connected to the EC2 instance using Session Manager.
10. Ran shell commands and exited the session.
11. Confirmed a log file appeared in S3 and downloaded the log for review.
12. Emptied the session-manager-logs bucket, removed the inline policy from the role, and deleted the CloudFormation stack to reduce costs.

## Inline Policy Pattern

---

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3::session-manager-logs-<redacted>/*"
    },
    {
      "Effect": "Allow",
      "Action": ["s3:GetEncryptionConfiguration"],
      "Resource": "*"
    }
  ]
}
```

The bucket ARN is redacted in evidence. The key points are the object-path suffix */\** for log writes and the encryption-configuration read permission needed by the logging workflow.

## Evidence

---

## Figure 2 - Session Manager S3 Logging Preferences Enabled

The screenshot shows the 'S3 logging' configuration page in the AWS IAM console. At the top, there is a toggle switch labeled 'enable' which is turned on. Below this, the 'S3 logging' section explains that session logs are stored and encrypted in an S3 bucket. The 'Send session logs to S3' option is checked and labeled 'Enable'. Under 'Enforce encryption', the 'Allow only encrypted S3 buckets' option is also checked. In the 'Choose S3 bucket' section, the 'Choose a bucket name from the list' radio button is selected, and a dropdown menu shows 'session-manager-logs' as the chosen bucket. A note indicates 'The bucket is encrypted'. At the bottom, the 'S3 key prefix - optional' field is empty, with a note stating 'To write output to a sub-folder, enter a sub-folder name.'

The preferences evidence shows Session Manager S3 logging enabled and the session-manager-logs bucket selected as the destination. Sensitive values are redacted.

## Figure 3 - SSMLogs Inline Policy JSON

### Specify permissions [info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

The screenshot shows the 'Policy editor' in the AWS IAM console, with the 'JSON' tab selected. The JSON editor displays the following code:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "s3:PutObject",
7       "Resource": "arn:aws:s3:::session-manager-logs-"
8     },
9     {
10      "Effect": "Allow",
11      "Action": [
12        "s3:GetEncryptionConfiguration"
13      ],
14      "Resource": "*"
15    }
16  ]
17 }
18
```

On the right side of the editor, there is a sidebar with tabs for 'Visual' and 'JSON'. Below these are sections for 'Edit statements', 'Add actions', 'Included' (showing 'S3'), 'Available' (listing various AWS services like 'API Gateway', 'ARC Region', etc.), 'Add a resource', and 'Add a condition'. At the bottom left of the editor, there is a button labeled '+ Add new statement'. The bottom right corner shows a timer '7:70' and the text 'JSON'.

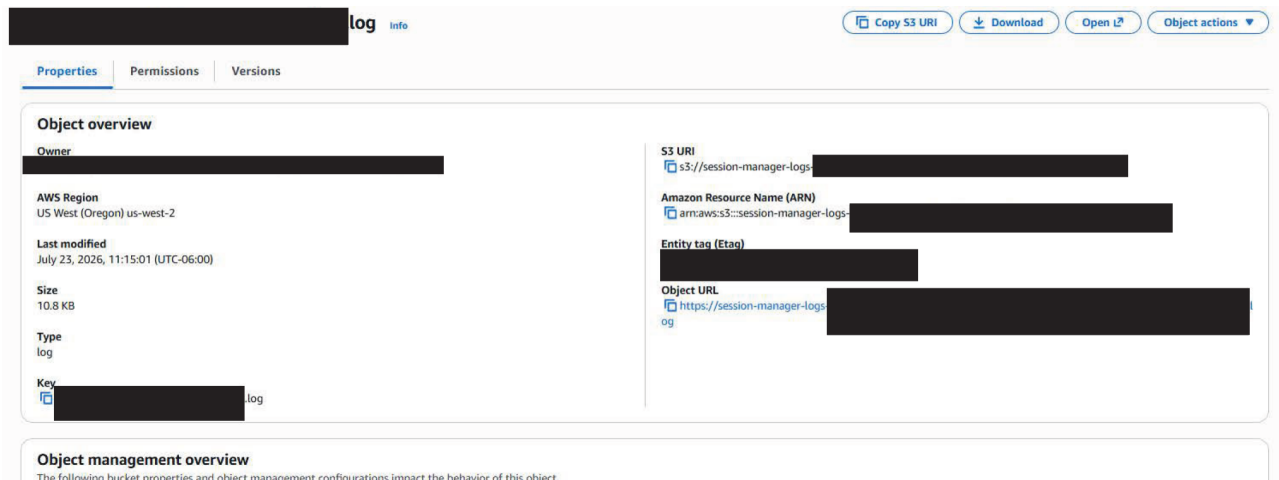
The inline policy evidence shows JSON granting s3:PutObject to the session-manager-logs object path and s3:GetEncryptionConfiguration. The bucket-specific ARN portion is redacted.

Figure 4 - SSMLogs Attached as Customer Inline Policy



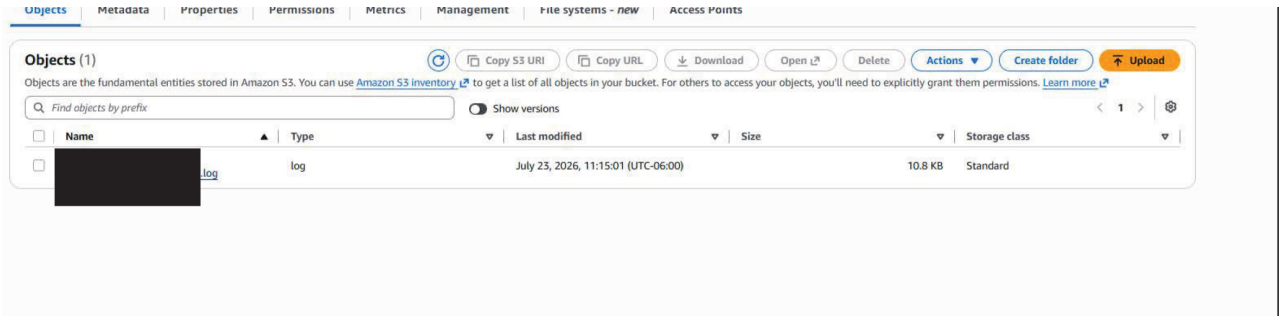
The IAM role evidence shows AmazonSSMManagedInstanceCore as an AWS managed policy and SSMLogs as a customer inline policy attached to the SSMClient role.

Figure 5 - Session Manager Login and Commands



The Session Manager evidence shows the browser-based session used to run commands before ending the session. Sensitive identifiers are redacted.

Figure 6 - S3 Session Log Object Confirmed



The S3 evidence shows the Session Manager log object available in the session-manager-logs bucket after the session ended. Sensitive details are redacted.

## Figure 7 - Downloaded Session Manager Log Excerpt

```
Script started on 2026-07-23 17:15:00+00:00 [<not executed on terminal>]
sh-5.2$

sh-5.2$ ls

['          clear          gawk          jsonpatch-3.9      msgen  ptx          semodule_link  systemd-c...
ac          clockdiff      gdbus          jsonpointer      msgexec  pwd          semodule_package
sy...
acpi_listen          cloud-id          gencat          jsonpointer-3      msgfilter  pwdx
      semodule...
addr2line          cloud-init          geqn          jsonpointer-3.9      msgfmt  pwmake          seq
sys...
alias          cloud-init-per          getconf          jsonschema          msggrep  pwscore          sestatus
...
amazon-ssm-agent      cmp          getent          kbd_mode      msginit  pydoc          setarch  syst...
apropos          col          getfacl          kbdrate      msgmerge  pydoc3          setfacl  systemd-d...
apropos.man-db      colcrt          getfattr          kbdinfo      msgunfmt  pydoc3.9
      setfattr...
ar          colrm          getkeycodes          kernel-install      msguniq  python3          setfont
sy...
arch          column          getopt          keyctl          mv  python3.9          setkeycodes  systemd-f...
arpaname          comm          getopts          kill          named-checkzone  pzstd          setleds  syst...
arping          command          gettext          killall          named-compilezone  quota
      setmetam...
as          coredumpctl          gettext.sh          kmod          named-nzd2nzf  quotasync          setpriv
...
at          cp          gio          last          namei  ranlib          setsid  systemd-machine-id-setup ...
atq          cpio          gio-querymodules-64  lastb          nano  read          setterm  systemd-mou...
atrm          cpupower          glib-compile-schemas  lastcomm          needs-restarting  readelf
      set...
attr          csh          gneqn          lastlog          neqn  readlink          setup-nsssysinit.sh
syste...
audit2allow          csplit          gnroff          lchfn          netstat  realpath
      setup-policy-ro...
audit2why          curl          gpasswd          lchsh          networkctl  recode-sr-latin          setvtrgb
...
austlast          cut          gpg          ld          newgidmap  red          sftp  systemd-run  whatis
austlastlog          cvtsudoers          gpg-error          ld.bfd          newgrp  rename          sg
systemd-...
austsyscall          cyrusdb2current          gpg2          ld.so          newuidmap  renew-dummy-cert  sh
...
austvirt          date          gpgme-json          ldd          ngettext  renice          sha1hmac  systemd-s...
awk          dbus-broker          gpgv2          less          nice  repo-graph          sha1sum  systemd-sy...
aws          dbus-broker-launch          gpic          lessecho          nisdomainname  repoclosure
      sha224...
```

The downloaded log excerpt shows the session start timestamp plus shell command activity, including the `ls` command and output.

## Security Analysis

This project adds resource-level activity logging to the private instance access pattern. CloudTrail records control-plane API calls, while Session Manager logs capture commands typed inside the interactive session.

The SSM Agent running on the instance needs permission to write log objects to S3. That is why the SSMClient role required the SSMLogs inline policy.

The S3 bucket destination keeps the log outside the instance, making it harder for an operator with shell access to remove the central copy of session activity.

The inline policy is scoped to the logging workflow: object write permission for the logging bucket and encryption-configuration read access.

This does not replace operating system logging. It records Session Manager command activity; system logs require a separate collection path.

Cleanup included infrastructure, bucket contents, and temporary permissions.

## Lessons Learned

Session Manager can store session activity logs in S3 or CloudWatch Logs.

The Session Manager preference defines the logging destination, but the instance role still needs S3 write permission.

Inline policies are useful for one-off role-specific permissions, but are not reusable like customer-managed policies.

The bucket object ARN must include the trailing /\* so the role can write objects inside the bucket.

Session logs capture commands typed during the session, not every operating system event.

Cleanup should include temporary permissions as well as infrastructure.

## Production Improvements

- Enable Session Manager logging centrally across workload accounts.
- Use customer-managed KMS keys for S3 log encryption when policy and cost requirements justify it.
- Send logs to a centralized logging account instead of a workload account bucket.
- Add bucket policies to prevent deletion or tampering by workload administrators.
- Enable S3 Object Lock or retention controls where required.
- Monitor StartSession, TerminateSession, and logging-preference changes with CloudTrail and EventBridge.
- Replace one-off inline policy edits with versioned Infrastructure as Code.

## Skills Demonstrated

| Session Manager logging | Amazon S3     | Inline IAM policy             |
|-------------------------|---------------|-------------------------------|
| SSMClient role          | s3:PutObject  | s3:GetEncryptionConfiguration |
| Command audit trail     | S3 log review | Policy cleanup                |

## Resume Bullet

### RESUME

Enabled AWS Systems Manager Session Manager S3 logging, configured the SSMClient role with an SSMLogs inline policy for encrypted S3 log delivery, validated captured session command output, and removed temporary permissions and infrastructure after testing.



# STAR Interview Story

---

**Situation:** Private EC2 access was available through Session Manager, but command activity was not retained for review.

**Task:** Enable Session Manager logging to S3 and validate that session activity was captured.

**Action:** I created the SLAW stack, enabled S3 logging, selected the session-manager-logs bucket, added SSMLogs to the SSMClient role, connected to the instance, ran commands, downloaded the log, cleaned the bucket, removed the policy, and deleted the stack.

**Result:** The lab demonstrated auditable private instance access and confirmed that Session Manager command activity could be captured in S3 while preserving cost-control cleanup.

## Version History

---

| v1.0.0 | July 23, 2026 | Initial documented implementation. Created SLAW stack, enabled Session Manager S3 logging, selected session-manager-logs bucket, added SSMLogs inline policy to SSMClient role, validated downloaded session log output, emptied bucket, removed policy, deleted stack, and embedded provided evidence using portfolio documentation standard v1.0.1. |
|--------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## References

---

- Cloud Security Lab a Week (S.L.A.W.) - Enabling Logs in Session Manager.
- User-provided Session Manager preferences screenshot.
- User-provided SSMLogs inline policy screenshot.
- User-provided IAM policy attachment screenshot.
- User-provided Session Manager login screenshot.
- User-provided S3 log object screenshot.
- User-provided downloaded SSM log file.
- Project 34 - Replace SSH with Session Manager Run As and CLI Access.